

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
!git clone https://github.com/Nishviprp/CodeGen-LLM.git
%cd /content/CodeGen-LLM
!pip install -q transformers peft accelerate datasets rouge-score sentencepiece tqdm matplotlib seaborn
!pip uninstall -y torchao
!pip install -q "peft==0.11.1"
```

```
Cloning into 'CodeGen-LLM'...
remote: Enumerating objects: 63, done.
remote: Counting objects: 100% (63/63), done.
remote: Compressing objects: 100% (54/54), done.
remote: Total 63 (delta 16), reused 35 (delta 7), pack-reused 0 (from 0)
Receiving objects: 100% (63/63), 13.73 MiB | 17.60 MiB/s, done.
Resolving deltas: 100% (16/16), done.
/content/CodeGen-LLM
  Preparing metadata (setup.py) ... done
  Building wheel for rouge-score (setup.py) ... done
Found existing installation: torchao 0.10.0
Uninstalling torchao-0.10.0:
  Successfully uninstalled torchao-0.10.0
  251.6/251.6 kB 21.9 MB/s eta 0:00:00
```

```
%cd /content/CodeGen-LLM
!python data/prepare_data.py
```

```
/content/CodeGen-LLM
Downloading CodeParrot dataset (5000 examples)...
This may take a few minutes on first run (downloads ~500MB).

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
README.md: 100% 1.30k/1.30k [00:00<00:00, 3.93MB/s]
Resolving data files: 100% 54/54 [00:00<00:00, 39700.69it/s]
  Collected 500/5000 examples...
  Collected 1000/5000 examples...
  Collected 1500/5000 examples...
  Collected 2000/5000 examples...
  Collected 2500/5000 examples...
  Collected 3000/5000 examples...
  Collected 3500/5000 examples...
  Collected 4000/5000 examples...
  Collected 4500/5000 examples...
  Collected 5000/5000 examples...

Done. Saved 5000 training examples to data/python_dataset.json
(Skipped 0 files that were too short)

Downloading HumanEval benchmark (164 tasks)...
README.md: 100% 6.52k/6.52k [00:00<00:00, 19.6MB/s]
openai_humaneval/test-00000-of-00001.par(...): 100% 83.9k/83.9k [00:01<00:00, 77.7kB/s]
Generating test split: 100% 164/164 [00:00<00:00, 5982.53 examples/s]
Done. Saved 164 HumanEval tasks to data/human_eval.json

All data ready. Next step: python models/finetune.py
```

```
%cd /content/CodeGen-LLM
import shutil, os
os.makedirs('/content/drive/MyDrive/CodeGenLLM_results', exist_ok=True)
!python evaluation/evaluator.py
# immediately copy results to Drive after it finishes
shutil.copytree('results', '/content/drive/MyDrive/CodeGenLLM_results', dirs_exist_ok=True)
print("Results copied to Google Drive safe even if this session ends.")
```

```
[108/164] HumanEval/107
[109/164] HumanEval/108
[110/164] HumanEval/109
[111/164] HumanEval/110
[112/164] HumanEval/111
[113/164] HumanEval/112
[114/164] HumanEval/113
[115/164] HumanEval/114
[116/164] HumanEval/115
[117/164] HumanEval/116
[118/164] HumanEval/117
[119/164] HumanEval/118
[120/164] HumanEval/119
[121/164] HumanEval/120
[122/164] HumanEval/121
[123/164] HumanEval/122
[124/164] HumanEval/123
[125/164] HumanEval/124
[126/164] HumanEval/125
[127/164] HumanEval/126
[128/164] HumanEval/127
[129/164] HumanEval/128
[130/164] HumanEval/129
[131/164] HumanEval/130
[132/164] HumanEval/131
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[137/164] HumanEval/136
[138/164] HumanEval/137
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[142/164] HumanEval/141
[143/164] HumanEval/142
[144/164] HumanEval/143
[145/164] HumanEval/144
[146/164] HumanEval/145
[147/164] HumanEval/146
[148/164] HumanEval/147
[149/164] HumanEval/148
[150/164] HumanEval/149
[151/164] HumanEval/150
[152/164] HumanEval/151
.....
```

```
%cd /content/CodeGen-LLM
!python evaluation/visualize.py
import shutil
shutil.copytree('results/visualizations', '/content/drive/MyDrive/CodeGenLLM_results/visualizations', dirs_exist_ok=True)
print("Charts copied to Google Drive.")
```

```
/content/CodeGen-LLM
Loading results...
Generating charts...
  Saved results/visualizations/pass_at_k.png
  Saved results/visualizations/latency.png
  Saved results/visualizations/complexity.png
  Saved results/visualizations/rouge_scores.png
  Saved results/visualizations/per_task_pass1.png

All charts saved to results/visualizations/
Charts copied to Google Drive.
```

```
from google.colab import files
import os
for f in os.listdir('/content/drive/MyDrive/CodeGenLLM_results'):
    full_path = f'/content/drive/MyDrive/CodeGenLLM_results/{f}'
    if os.path.isfile(full_path):
        files.download(full_path)
```

Start coding or [generate](#) with AI.

